IOP Publishing Biofabrication 16 (2024) 045014 https://doi.org/10.1088/1758-5090/ad6374

Biofabrication

RECEIVED
10 May 2024

REVISED
15 June 2024

ACCEPTED FOR PUBLICATION
15 July 2024

PUBLISHED
25 July 2024

PAPER

A practical machine learning approach for predicting the quality of 3D (bio)printed scaffolds

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Keywords: machine learning, deep learning, clustering, 3D printing, bioprinting, tissue engineering

Supplementary material for this article is available online

Abstract

3D (Bio)printing is a highly effective method for fabricating tissue engineering scaffolds, renowned for their exceptional precision and control. Artificial intelligence (AI) has become a crucial technology in this field, capable of learning and replicating complex patterns that surpass human capabilities. However, the integration of AI in tissue engineering is often hampered by the lack of comprehensive and reliable data. This study addresses these challenges by providing one of the most extensive datasets on 3D-printed scaffolds. It provides the most comprehensive open-source dataset and employs various AI techniques, from unsupervised to supervised learning. This dataset includes detailed information on 1171 scaffolds, featuring a variety of biomaterials and concentrations—including 60 biomaterials such as natural and synthesized biomaterials, crosslinkers, enzymes, etc.—along with 49 cell lines, cell densities, and different printing conditions. We used over 40 machine learning and deep learning algorithms, tuning their hyperparameters to reveal hidden patterns and predict cell response, printability, and scaffold quality. The clustering analysis using KMeans identified five distinct ones. In classification tasks, algorithms such as XGBoost, Gradient Boosting, Extra Trees Classifier, Random Forest Classifier, and LightGBM demonstrated superior performance, achieving higher accuracy and F1 scores. A fully connected neural network with six hidden layers from scratch was developed, precisely tuning its hyperparameters for accurate predictions. The developed dataset and the associated code are publicly available on https://github.com/saeedrafieyan/MLATE to promote future research.

1. Introduction

In recent years, the field of tissue engineering has undergone a revolution with the widespread adoption of 3D printing. This technology has increasingly facilitated the fabrication of complex structures characterized by their intricate configurational features [1, 2]. The ability to incorporate cells directly into scaffolds during fabrication, a process known as 3D bioprinting, stands out as a unique feature that is nearly impossible to achieve with other scaffold fabrication methods. Additionally, its specific design capabilities, on-demand fabrication, low cost, and the potential for high structural complexity make it a highly desirable method in tissue engineering [3, 4]. 3D printing technology has vast applications across multiple areas of tissue engineering, including the fabrication of tissues such as skin, cartilage, liver, lungs, dental, neuronal tissue, and the pancreas [4–7]. Various materials, including natural polymers such as alginate, fibrin, collagen, gelatin, chitosan, and hyaluronic acid (HA), as well as biodegradable synthetic polymers like thermoplastic polyesters (e.g, polylactic acid (PLA), polylactic-co-glycolic acid (PLGA), polycaprolactone (PCL), and polyethylene glycol (PEG)), are extensively utilized in biomedical 3D printing [8–10].

Artificial intelligence (AI) aims to equip machines with the capability to mimic human cognition. Its applications are expanding across various fields today, including biomedical engineering. AI is a

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general term encompassing multiple areas utilizing predictive statistical models [11]. Machine learning (ML), a subset of AI, comprises diverse techniques enabling computers to learn and make informed decisions [12, 13]. ML is categorized into supervised learning, where models are trained on labeled data to make accurate predictions, and unsupervised learning, which uncovers hidden structures and patterns in unlabeled data [13]. Deep learning (DL) is a subset of ML involving neural networks with many hidden layers; 'deep' refers to these hidden layers [14, 15].

In the contemporary landscape of biomedical research, AI has emerged as a transformative force across various domains of regenerative medicine, encompassing drug discovery, disease modeling, predictive modeling, personalized medicine, tissue engineering, cell therapy, clinical trial design, patient monitoring, patient education, and regulatory compliance [16–20]. Our examination specifically narrows to the integration of AI within tissue engineering. This multidisciplinary field, bridging engineering, biology, and medicine, represents an innovative frontier aimed at devising groundbreaking methods for disease treatment that are not cured in traditional ways [21].

In tissue engineering, scaffolds are pivotal components, serving as foundational platforms that facilitate cell growth and proliferation. The core objective of this discipline is to mimic organ structures in a laboratory setting by utilizing biomaterials. Researchers engage in the meticulous process of crafting these scaffolds, subsequently seeding cells onto them to observe cellular behaviors and responses. The outcomes of these cellular interactions serve as critical indicators, guiding scientists in assessing the efficacy and quality of the developed scaffolds [12, 21, 22].

Scaffold fabrication in tissue engineering is crucial for replicating organ structures, with techniques aimed at producing repeatable outcomes and offering control over various parameters [23]. Among these methods are hydrogel formation, electrospinning, freeze-drying, and, notably, 3D (bio)printing, which falls under the broader category of additive manufacturing (AM). AM encompasses several techniques, including Fused Deposition Modeling, inkjet bioprinting, light-assisted bioprinting, and extrusion-based bioprinting. Extrusion-based bioprinting is particularly prominent due to its versatility in handling diverse materials and ease of use [15, 24]. Given the intricate challenge of replicating tissue, which involves numerous variables requiring precise adjustment, and the capacity of AI to discern complex relationships among various parameters, numerous researchers have explored diverse applications of AI within the 3D (bio)printing process [25].

Tian *et al* provided two datasets for their study, which comprised 617 cell viability instances and another comprising 339 filament diameter instances [26]. They tackled regression and classification tasks, categorizing cell viability, filament diameter, and pressure into binary classes (acceptable or not) based on predefined thresholds. They employed random forest, logistic regression, and support vector machines for the classification tasks to predict binary classes for cell viability, filament diameter, and pressure. The regression tasks aimed to predict the continuous values of cell viability, filament diameter, and pressure by support vector machines, linear regression, and random forest. Their dataset contained material compositions, cell density, and printing conditions [26]. After thoroughly examining their dataset, our analysis reveals that the number of unique scaffolds in their cell viability dataset represents approximately 50% of the reported instances. This deficiency arises from multiple cell viability measurements reported for each scaffold at different observation times. Bone *et al* conducted a study with a compact dataset of 48 scaffold samples, incorporating system parameters such as normalized flow rate, alginate concentration, nozzle speed, and nozzle diameter as inputs. Their output was the print score. They employed a neural network architecture featuring four hidden layers to analyze this data. This approach yielded an R^2 value of 0.643 [27]. Fu *et al* developed a dataset comprising 12 data points aimed at predicting the printability of Pluronic F127. This dataset incorporated input variables such as biomaterial concentration, nozzle temperature, and printing path height. Utilizing a support vector machine for their binary classification analysis achieved an accuracy of 83.3% [28]. Conev *et al* applied a dataset featuring 72 printed specimens—comprising poly(propylene fumarate) concentration, print-related characteristics, and design considerations—to train random forest models via regression and binary classification for predicting print quality [29]. Ruberu *et al* provided a dataset of 177 samples for assessing printability and expediting optimization for extrusion-based 3D bioprinting, targeting consistent fabrication of 3D scaffolds. Their research employs Bayesian Optimization to devise a scoring scheme evaluating the printability of gelatin methacryloyl (GelMA) and HA methacrylate (HAMA) bioinks [30]. Li *et al* used a three-layer backpropagation artificial neural network on a dataset with 9 data to predict scaffold deformation [31]. Chen *et al* collected the information from 210 3D printed scaffolds and performed binary classification using decision tree, random forest, and DL to predict biomaterials printability [32].

Despite progress in the field, some crucial limitations dimmed the wide usage of AI in tissue engineering. First, data is an essential element in AI applications, with the quality and quantity of data determining factors in a trained model's ability to be used in real-world scenarios. Many researchers do not make their datasets publicly available, and a lack of data has hindered the proliferation of AI and data-driven research in this field. Based on Bonatti *et al*, only about a quarter of published papers on using ML in bioprinting have made their dataset publicly available [15].

Second, ML and DL algorithms need a lot of data, especially DL algorithms. Based on our literature review, many researchers have implemented DL algorithms on a minimal dataset [33]. Using small datasets could cause overfitting (when an ML model performs well on the training data but poorly on new, unseen data because it has learned noise or random fluctuations in the training data rather than the underlying patterns). Although their models achieved high scores in evaluation metrics, they could not generalize and were unsuitable for actual usage. Most models trained under these circumstances cannot even correctly predict the result of a data point out of previously seen data.

Third, most works in the field rely on popular algorithms and techniques, which may not always represent best practices. Based on the literature review, we found that most researchers tend to use simpler methods like binary classification.

Fourth, as we mentioned earlier, most works do not have a practical view and do not develop models that other researchers could use widely. Most developed models are limited to tiny amounts of biomaterials and include some of the most used biomaterials, like alginate and gelatin. Some researchers used some features as inputs that were not controllable before manufacturing.

Additionally, some trained models only apply under specific research conditions for which they were trained, limiting their use by other researchers. The purpose of conducting AI projects is to develop models on a wide range of data and conditions to give the model the capacity to predict in real-world usage, which could assist scientists during their research. This goal cannot be achieved if the most suitable features are not chosen during the development. Therefore, to develop a reliable ML model to predict the results, we should consider having a considerable amount of accurate and quality data with various useful features.

To address these challenges, the most comprehensive publicly available dataset of 3D (bio)printed scaffolds, including the information of 1171 ones, is provided in this work. This dataset contains 60 different materials and 49 cell lines, including 6 co-cultured systems. Various conventional and cutting-edge algorithms were applied, including clustering, ML, and DL algorithms. We used more than 40 algorithms and made all our codes and datasets publicly available for everyone to speed up the progress of AI in tissue engineering. With this dataset and public codes, we could expect other applications such as algorithmically optimized scaffolds, AI-designed scaffolds, or generative AI for copilot development, which will be developed by different researchers soon.

2. Methods

2.1. Data gathering

For data gathering, we utilized various resources, including datasets provided in the literature, such as those by Tian *et al* [26] and Ruberu *et al* [30]. Additionally, we incorporated data from other research studies that employed 3D (bio)printing for fabricating tissue engineering scaffolds by searching Scopus using 'tissue engineering,' 'bioprinting,' and '3D printing' as keywords. We filtered the papers based on the completeness of the information they provided. Papers that did not specify printing conditions (such as pressure, temperature, nozzle diameter, etc), scaffold properties (such as materials and material compositions, etc), or details on cell lines and cell densities (for bioprinting studies) were excluded. Additionally, papers that provided this information in a vague manner that prevented data extraction were also excluded.

2.2. Dataset creation

The initial step involved verifying the accuracy of the data collected from datasets available in the literature. When utilizing public datasets, we started by conducting a thorough review of all the data. We discovered that some materials used in the samples had been overlooked, certain compositions were incorrect, and some rows of data needed to be removed.

The next step was identifying the essential features required to create a dataset that meets our specifications. We identified and collected data from the literature using the controllable features that a researcher employs before initiating the printing process. Furthermore, if certain essential features were absent from public datasets, we either added them or removed irrelevant ones.

For instance, in the dataset provided by Tian *et al*, we removed features irrelevant to our study, including those not known before manufacturing, such as Fiber Diameter (μm), Fiber Spacing (μm), Viability at the Time of Observation (%), and various categorical features such as DI Water Used?, Precrosslinking Solution Used?, Saline Solution Used?, EtOH Solution Used?, Photoinitiator Used?, Enzymatic Crosslinker Used?, Matrigel Used?, Acceptable Viability (Yes/No), and Acceptable Pressure (Yes/No). We also removed the Days Observed column to ensure the uniqueness of the samples. In the Tian *et al* dataset, if a sample was observed on different days and cell viability was reported for each observed time, it was considered a unique sample. Removing these columns resulted in many duplicate samples, which we subsequently eliminated. This dataset did not contain 3D-printed samples and cell lines; some bioprinted samples were overlooked. Therefore, we gathered these samples from the original papers.

After treating these datasets, we further enhanced our samples by extracting additional data from relevant research papers.

2.3. Data pre-processing

The next step involved ensuring that the units for each column were consistent. Additionally, we verified the accuracy of all values entered in each column. Numeric values are essential when employing ML algorithms. However, the reported data was sometimes qualitative rather than quantitative or presented as a range instead of an exact value. After correcting these discrepancies, the subsequent step involved addressing missing data. Some missing data could be computed by leveraging certain relevant values, while others were not reported in the original paper source.

Only the cell line column was categorical in the provided dataset, so we used label encoding to convert these entries into numerical values. We employed the Iterative Imputer with XGBoost regression as the estimator for imputing missing values, using 1000 iterations to predict the missing values. Additionally, for data reported as a range, if we could not determine the exact numbers from the article text or its supplementary materials, we used the mean to convert the range to a single value.

2.4. Sample rating

Since both 3D printed and 3D bioprinted samples were used, we had to create a rating system that could cover both. After preparing the data, they had to be rated by our criteria. The goal was to predict the scaffold quality for tissue engineering applications and its dependence on the quality of the scaffold and the response of cells cultured on them. Therefore, the following criteria were defined.

A) **Cell Response**: a rating method from 1 to 5 determines cell behavior (table 1).
B) **Printability:** a rating method from 0 to 3 that determines the print quality (table 2).
C) **Scaffold quality**, shown in equation (1), which is the multiplication of cell response by printability, serves as a criterion for evaluating the goodness of a scaffold.

$$\text{Scaffold Quality} = \text{Cell Response} \times \text{Printability}. \quad (1)$$

2.5. Computational resources

We used an Asus TUF F15 laptop equipped with an NVIDIA GeForce RTX 3070 GPU for all analyses in this research. The analyses were performed using the Python programming language. The following frameworks and libraries were utilized for analysis: PyTorch, Scikit-learn, Pandas, NumPy, Plotly, and Matplotlib.

Table 1. The details of the cell responses rating system.

Label	Cell Response
1	Bioprinting did not occur.
2	The cell response was inadequate in the short term (up to three days). Cell viability, growth, proliferation, and differentiation were not good, and many dead cells were observed in the live/dead or MTT test for the scaffolds.
3	Good cell response in short term (up to three days)
4	Good cell response in the short term (up to three days) but bad in the long term (more than 3 d)
5	Good cell response in the short and long term (more than three days)

Table 2. The details of the printability rating system.

Label	Printability
0	The ink was not extruded.
1	The ink behaved like a liquid, formed beads, etc.
2	The ink was extrudable but was not optimized.
3	The ink was extrudable and optimized.

2.6. Data analysis

The prepared dataset contains 1171 samples with 72 input and 3 output columns. Among the 72 input columns, 60 correspond to biomaterials and their concentrations, 2 columns are dedicated to cell line and cell density, 2 columns cover physical and photo crosslinking times, and 8 columns pertain to printing conditions. A thorough analysis was conducted by visualizing various aspects of the data to gain a deeper insight into our dataset. It includes exploring the distribution of data points, the diversity of materials used, and other relevant parameters. By graphically representing these elements, we could identify the ranges of different parameters and pinpoint the most frequently utilized materials and cell lines, among other findings. This visualization enhanced the understanding of the dataset's characteristics, facilitating a more informed approach to subsequent analyses and modeling efforts.

2.7. ML techniques

In our analysis, to find the algorithm that performs better on the presented dataset, a mix of supervised and unsupervised learning algorithms was systematically employed to delve into the intricacies of the dataset. Clustering algorithms were used to uncover hidden patterns within the data, highlighting inherent groupings and relationships. Additionally, ML and DL algorithms were applied for tasks such as classifying cell responses, printability, and the quality of scaffolds. The primary objective of this phase was to identify the most effective algorithm for each specific task, ensuring precision and reliability in our predictions and analyses. Because our labels are discrete, we focused on classification tasks suited for discrete targets. Regression tasks are typically used for continuous targets, which did not apply to our dataset. It is worth noting that we used oversampling methods for supervised tasks to balance our dataset labels.

2.8. Hyperparameter tuning

To optimize the performance of ML algorithms, we identified the best hyperparameters using a grid search approach with 10-fold cross-validation, and for DL algorithms, we employed Optuna [34], both conducted exclusively on the training set. This meticulous process ensured that the models were optimally tuned for maximum accuracy and efficiency. The ideal hyperparameters were then applied to the testing set to evaluate the models' performance under real-world conditions.

2.9. Data evaluation

To compare the performance of various algorithms used for classification, we employed seven key metrics: precision, recall, accuracy, F1-score, Area Under the Receiver Operating Characteristic Curve (AUC), Cohen's kappa coefficient, and Matthews Correlation Coefficient. These metrics, detailed in a previous study [12], offer distinct insights into the algorithms' effectiveness:

- **Precision:** Evaluate the accuracy of positive predictions by calculating the ratio of true positive results to the total predicted positives [12, 35].

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

- **Recall:** Assesses the algorithm's ability to identify all actual positives by calculating the ratio of true positives to the total actual positives [12, 35].

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

- **Accuracy:** Measures the overall correctness of predictions by calculating the ratio of correct predictions to the total number of cases. Notably, accuracy could be misleading in an imbalanced dataset [12].

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Totall number of predictions}} \tag{4}$$

- **F1-score:** Provides a harmonized measure of precision and recall by calculating their harmonic mean.

$$F1\ \text{Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})} \tag{5}$$

- **Area Under the Receiver Operating Characteristic Curve (AUC):** Reflects the model's capacity to differentiate between classes by plotting the true positive rate against the false positive rate at various threshold settings. A higher AUC indicates better model performance [12].
- **Cohen's kappa coefficient:** Quantifies the agreement between predicted and actual classifications, correcting for random chance. It ranges from -1 to 1, with higher values indicating better agreement [12, 36].

$$\kappa = \frac{\text{Actual agreements beyond chance}}{\text{Potential agreements beyond chance}} \tag{6}$$

- **Matthews Correlation Coefficient (MCC):** Provides a balanced measure of the true and false positives and negatives. It is ideal for evaluating classification performance, especially with imbalanced classes, and ranges from -1 to 1, with higher values indicating better performance [12, 37].

$$\text{MCC} = \frac{(\text{TP} \times \text{TN}) - (\text{FP} \times \text{FN})}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}}. \tag{7}$$

TP, TN, FP, and FN stand for True Positives, True Negatives, False Positives, and False Negatives, respectively. This comprehensive evaluation framework allows for an in-depth analysis of each algorithm's performance, highlighting both its strengths and limitations.

For clustering algorithms, we used the following metrics to evaluate the quality of the clusters:

- The **Silhouette Score** measures how similar an object is to its cluster (cohesion) compared to other clusters (separation). It ranges from -1 to 1, where higher values indicate that objects are well-matched to their cluster and poorly matched to neighboring clusters. High values across most objects suggest an appropriate clustering configuration [38, 39].
- The **Davies-Bouldin Score** is calculated by averaging the similarity between each cluster and its most similar cluster, determined by the ratio of distances within clusters to distances between clusters. Farther apart and less dispersed clusters result in a lower score, with zero being the ideal score indicating optimal clustering [40].
- The **Calinski and Harabasz score** is defined as the ratio of the dispersion between clusters to the dispersion within clusters. A higher score indicates better-defined clusters [41].

These metrics provide a comprehensive understanding of the clustering performance, enabling us to assess the quality and appropriateness of the clusters formed by the algorithms.

3. Results

3.1. Dataset creation

We gathered data for both 3D printed and 3D bioprinted scaffolds, focusing on variables such as biomaterial types, compositions, cell lines, and cell density (specifically for bioprinting). Additionally, printing conditions were considered, including extrusion pressure, nozzle speed, nozzle temperature, and substrate temperature. In line with our practical approach, we opted not to include features such as porosity, pore size, tensile strength, compressive strength, Young's modulus, fiber diameter, and volumetric extrusion rate. Although these parameters are essential, they are either uncontrollable or challenging to control before printing begins. Additionally, the influence of these parameters can be indirectly observed through cell responses [12]. Our selection criteria were guided by ensuring the models' practical applicability and reproducibility, focusing on parameters that can be precisely managed and adjusted during the preparation phase of the printing process.

3.2. Dataset visualization

To gain deeper insights into the dataset's characteristics, we began our analysis by visualizing its various dimensions. This essential first step helped us uncover patterns, distributions, and connections within the data, setting the stage for more detailed analytical and modeling efforts. Our dataset is structured into four primary categories: biomaterials, cell lines and densities, printing conditions, and targets such as cell response, printability, and scaffold quality.

Figures 1 and 2 depict our analysis, presenting a comprehensive overview of the given dataset specifications, including the frequencies of all four primary categories. It details the percentages of scaffolds manufactured through 3D printing and 3D bioprinting, along with the frequencies of target outcomes. Figures 2(A) and (B) also explore different cell lines' diverse usage frequencies and densities. Additionally, the heatmap in figure 1(B) illustrates the co-usage of materials, helping to identify which materials are most commonly used together.

Additionally, table S1 provides a statistical analysis of the biomaterials included in the dataset. For each biomaterial, it lists units of measurement and presents a comprehensive statistical summary. This summary includes the frequency of use, minimum, mean, median, maximum values, and the standard deviation for each biomaterial. This quantitative overview is crucial for understanding usage patterns' variability and central tendency, facilitating a more informed selection of materials for specific applications.

3.3. ML techniques

Before implementing the ML algorithms, addressing the issue of missing data is essential. For the imputation of these missing values, we utilized XGBoost regression. Furthermore, given the high-dimensional nature of our dataset, which comprises 75 columns—72 as input features and 3 output features—dimensionality reduction is necessary for adequate visualization. Therefore, we applied dimensionality reduction techniques to condense the dataset to a two-dimensional space, thus facilitating clearer graphical representation and interpretation. Additionally, it is essential to note that for each analytical task, various algorithms were evaluated to ascertain the most effective model that outperforms others in the context of our target.

3.3.1. Clustering

As clustering algorithms are unsupervised, they do not require labeled data. Therefore, the three label columns from our dataset were dropped to implement them. We employed 11 different clustering algorithms to cluster the presented dataset and evaluated them using three metrics: the Silhouette Score, Davies-Bouldin Score, and Calinski-Harabasz Score. Several tests were conducted to determine the appropriate number of clusters, considering two main factors: the need for a reasonable number of clusters and, as outlined in section 2.9, the satisfactory performance according to the evaluation metrics for this chosen number of clusters. We utilized the Silhouette Score analysis, Davies-Bouldin Score analysis, Calinski-Harabasz Score analysis, and the elbow method to identify the optimal number of clusters. The elbow method involves plotting the explained variance as a function of the number of clusters and identifying where the increase in variance explained becomes negligible, resembling an 'elbow'. As depicted in figure 3, these analyses indicated that the optimal number of clusters is approximately five. We then tested various algorithms to determine which performed best within this range of clusters. It is important to note that some clustering algorithms do not permit direct specification of the number of clusters. The results are detailed in table S2, and figure 4 illustrates the clustering outcomes of the data distribution.

To determine the algorithm with superior performance, we considered assessment criteria such as the number of clusters, cluster density, and balance. After reviewing the results and discussing evaluation metrics in section 2.9, it became evident that k-means outperformed other algorithms. Consequently, we selected k-means for our analysis due to its good performance, suitability in the number of clusters, and favorable outcomes on evaluation metrics. Additionally, the data distribution within the k-means clusters is more evenly balanced.

After selecting the clustering algorithm, we analyzed the data to identify the differences between clusters. We performed a comprehensive analysis

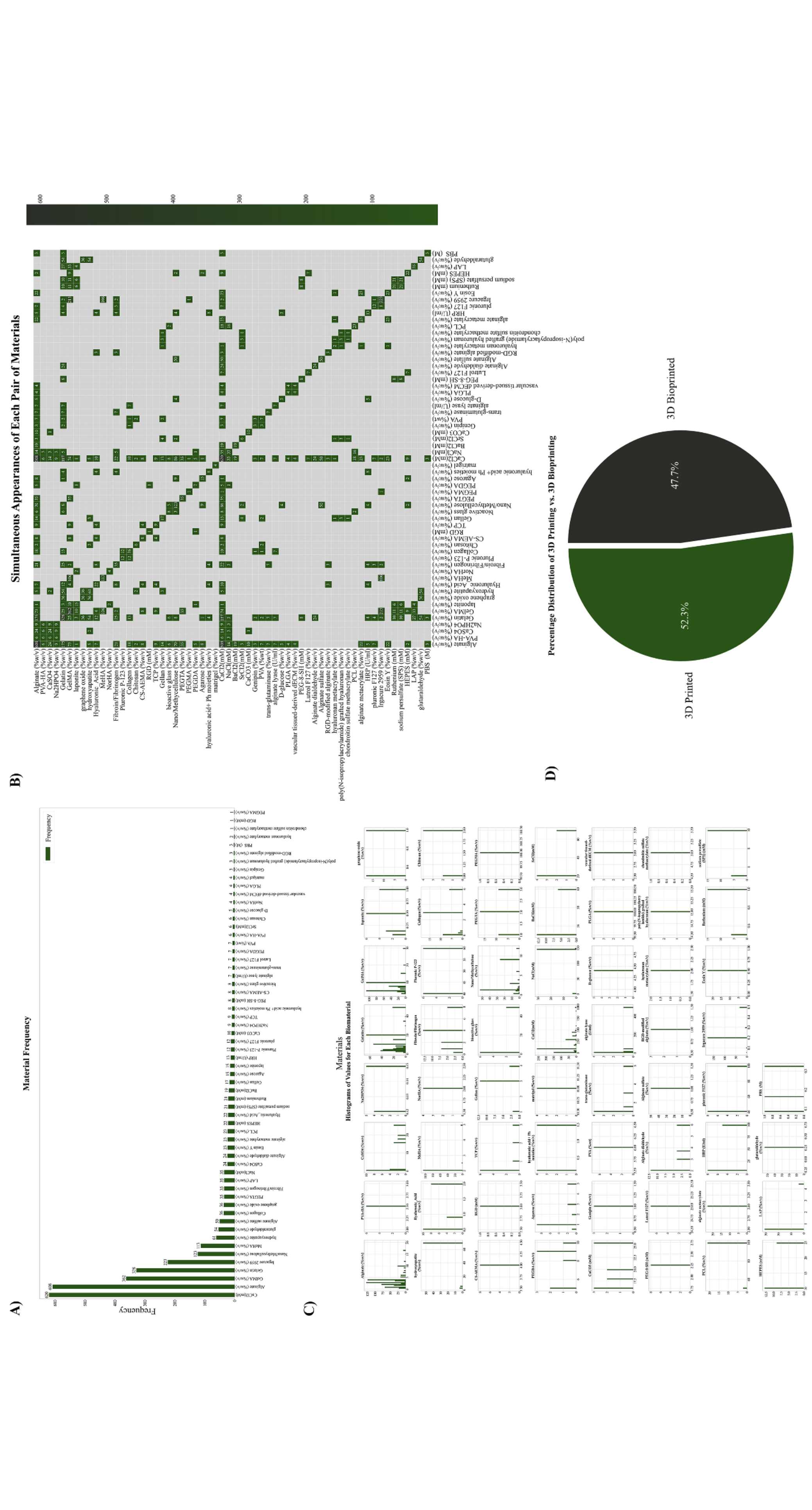

Figure 1. Multifaceted Analysis of the Biomedical Engineering Dataset. (A) Material Frequency: Bar graph displaying the frequency of material usage across different scaffolds within the dataset. (B) Simultaneous Appearances of Each Pair of Materials: Heatmap visualizing the co-occurrence of material pairs, highlighting the most commonly used combinations. (C) Histograms of Values for Each Biomaterial: A series of histograms representing the distribution of each biomaterial. (D) Percentage Distribution of 3D Printing vs. 3D Bioprinting: Pie chart illustrating the proportion of scaffolds produced by 3D printing versus 3D bioprinting methods.

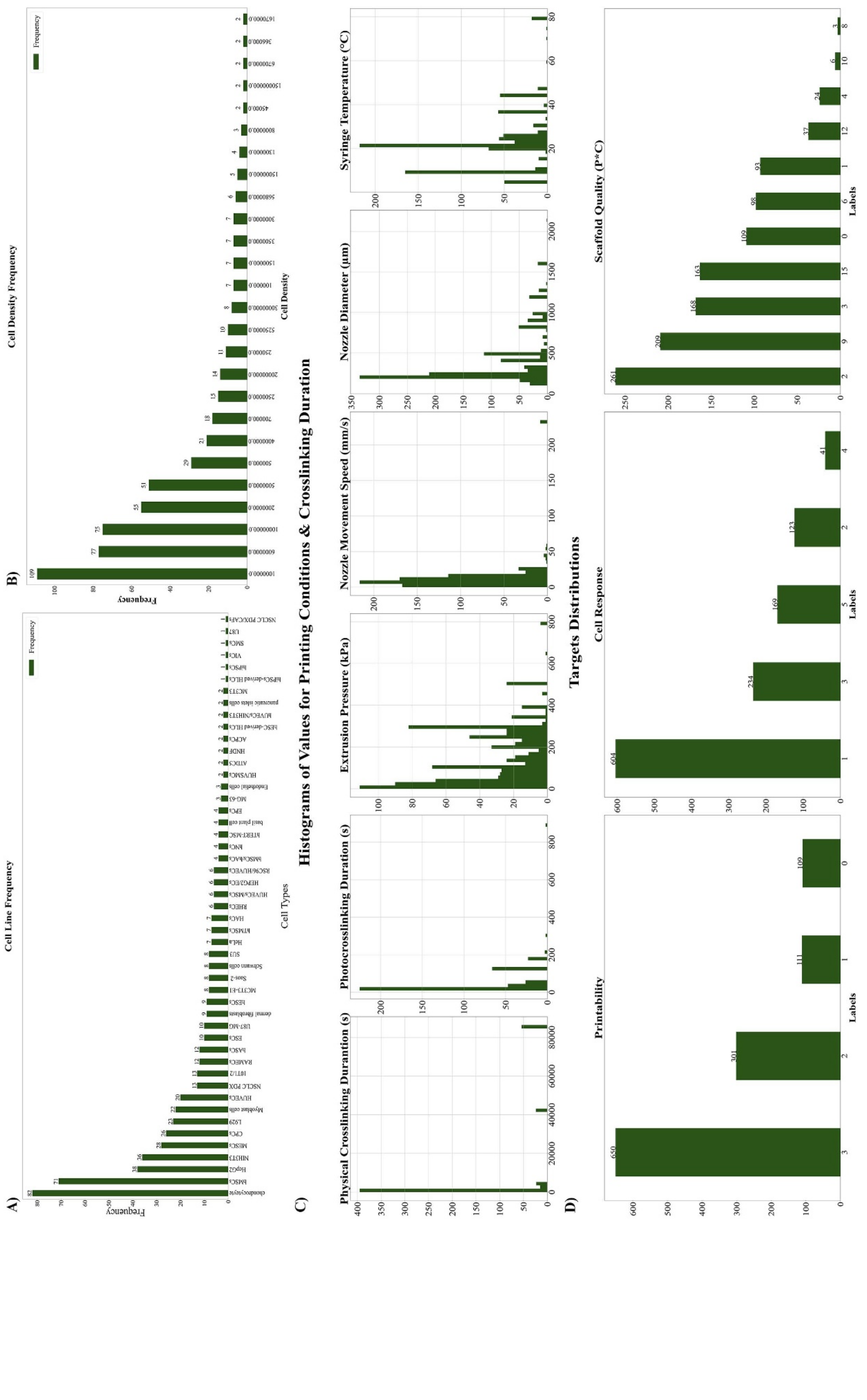

Figure 2. (A) Cell Type Frequency: Bar graph indicating the frequency of different cell types used in the scaffolds. (B) Cell Density Frequency: Bar graph showing the frequency distribution of cell densities across the scaffolds. (C) Histograms of Values for Printing Conditions and crosslinking durations. (D) Distributions of different targets in printability, cell response, and scaffold quality.

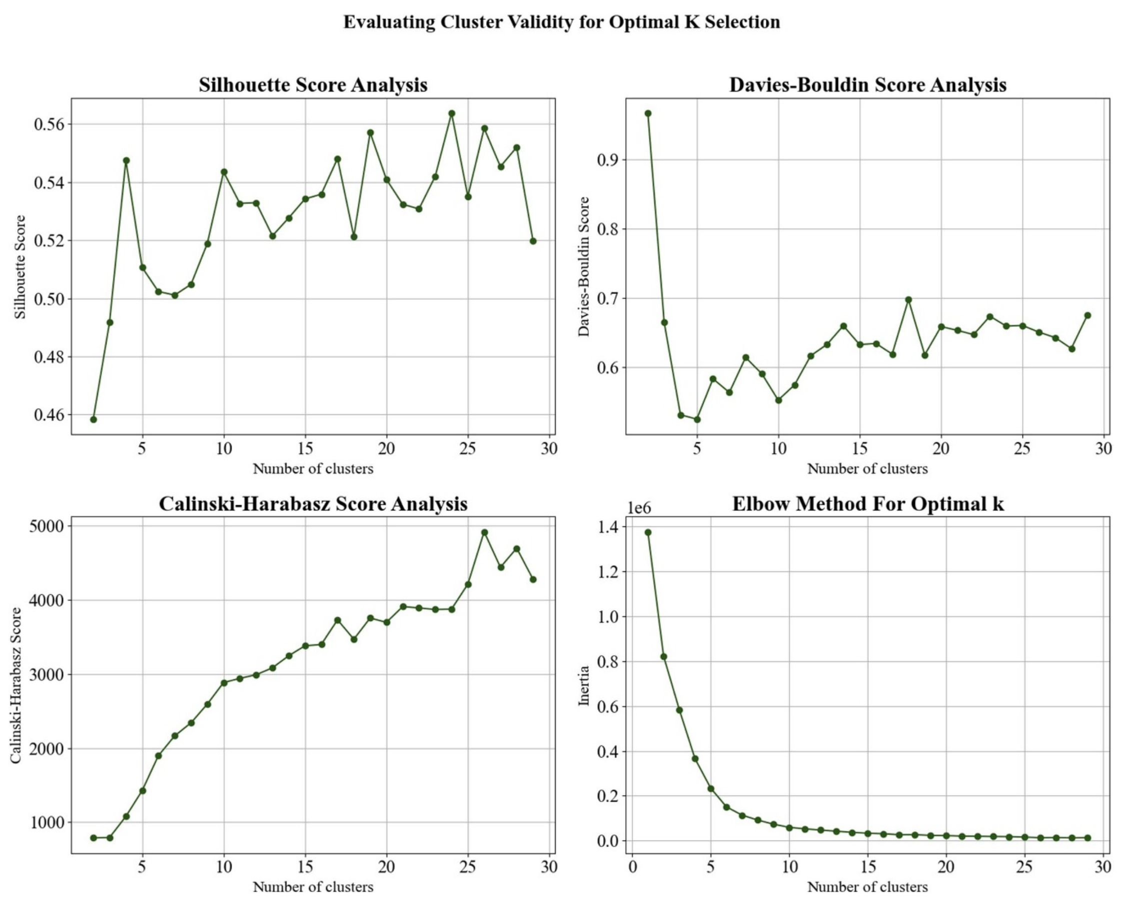

Figure 3. Tests to find the optimal number of clusters.

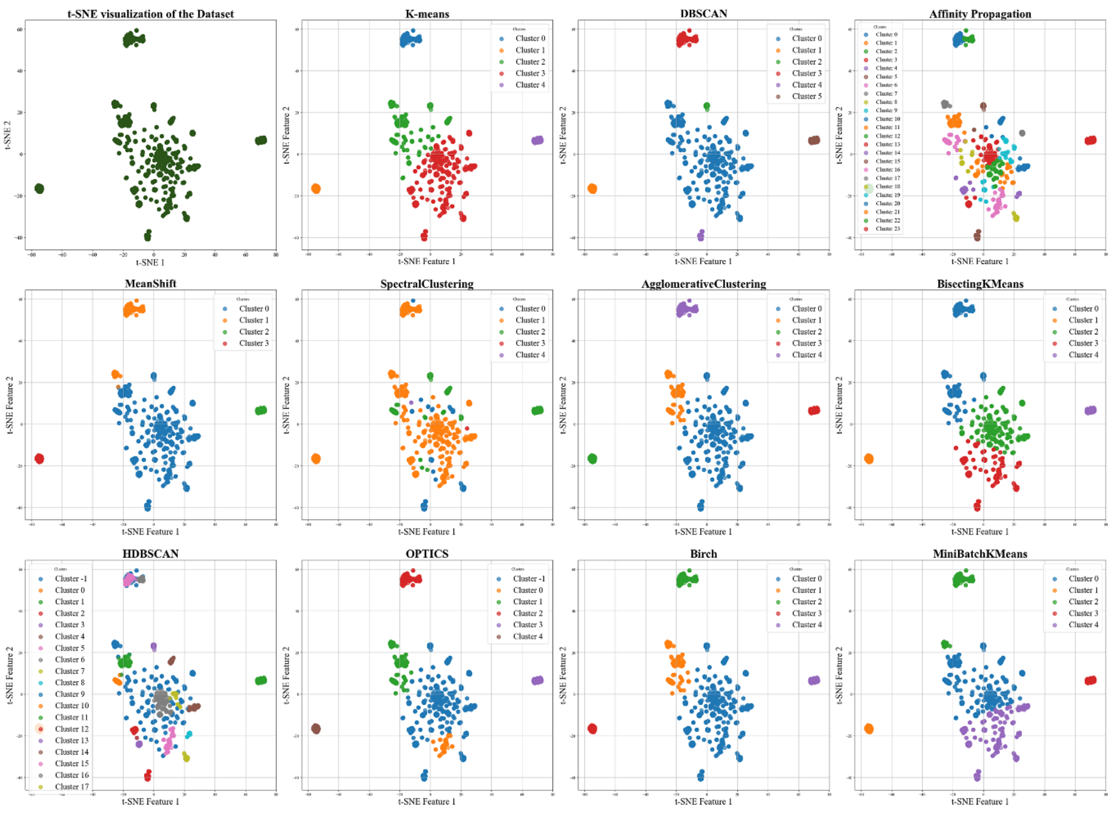

Figure 4. The visualization of data distribution on 2D space and clustering results.

Table 3. Examination of each cluster based on materials.

Clusters	Materials
0	This cluster just contains: GelMA (%w/v), MeHA (%w/v), Irgacure 2959 (%w/v)
1	This cluster just contains: Nano/Methycellulose (%w/v), CaCl2(mM), Alginate sulfate (%w/v)
2	The top five materials in this cluster are: GelMA (%w/v), Irgacure 2959 (%w/v), CaCl2(mM), Gelatin (%w/v), Alginate (%w/v)
3	The top five materials in this cluster are Alginate (%w/v), CaCl2(mM), Gelatin (%w/v), Nano/Methycellulose (%w/v), GelMA (%w/v)
4	This cluster just contains: Gelatin (%w/v), graphene oxide (%w/v), hydroxyapatite (%w/v), glutaraldehyde (%w/v)

Table 4. Examination of each cluster based on cell lines.

Clusters	Cell line
0	This cluster contains NIH3T3.
1	This cluster just contains chondrocyte cells.
2	This cluster's top five used cell lines are bMSCs, NIH3T3, HepG2, HUVECs, and HACs.
3	This cluster's top five cell lines are bMSCs, chondrocytes, MESCs, CPCs, L929, HepG2, and Myoblast cells.
4	This cluster does not contain any bioprinted scaffold; therefore, it does not contain any cell line.

and provided interactive charts in the supplementary materials to enhance understanding of the characteristics of each cluster. In tables 3 and 4, we present a brief exploration of the materials and cell lines for each cluster, respectively.

3.3.2. Classification

Our dataset features three distinct targets, each with its class distribution, presented in figure 2(D). This figure reveals that the class distributions are unbalanced across these targets. To address this imbalance before proceeding with the classification tasks, we implemented oversampling techniques tailored to each target class to achieve a balanced dataset. Figure 5 confirms that the data distribution before and after oversampling remains consistent, indicating that the methods did not negatively affect the data quality. Table S3 examines the number of data points per class for each target before and after the oversampling process.

3.3.2.1. ML algorithms

We conducted hyperparameter tuning through grid search across 29 algorithms to identify the one with the optimal hyperparameters for superior performance on presented datasets, focusing on predicting cell response, printability, and scaffold quality. This extensive process involved over 56 000 fits to pinpoint each target's most effective algorithm and hyperparameters. Our method underscored the necessity of evaluating numerous algorithms with varying parameters to discover the most suitable one.

As depicted in figure 6, for predicting Scaffold Quality, XGBoost, LightGBM, and Gradient Boosting Classifier performed best with F1 scores of 0.88, 0.87, and 0.86, respectively. For Cell Response prediction, the Gradient Boosting Classifier led with an F1 score of 0.83, closely followed by the Random Forest and Extra Trees Classifier at 0.82, performing nearly identically. Considering printability, Extra Trees Classifier, LightGBM, and XGBoost topped the list with equal F1 scores of 0.96, while Gradient Boosting Classifier and Random Forest also matched each other with scores of 0.95. The third tier included the Bagging Classifier and Hist Gradient Boosting Classifier, with an F1 score of 0.94. The detailed results of tuning all algorithms for Scaffold Quality, Cell Response, and Printability are presented in table S4. In figure 7, the ROC curves for all 29 algorithms are collectively displayed in separate plots for Scaffold Quality, Cell Response, and Printability, facilitating a comprehensive comparison of their performance. Additionally, for enhanced visualization, we have plotted the ROC curves individually for each target in figures S1–S3.

3.3.2.2. DL algorithms

We implemented DL algorithms using our oversampled datasets to ensure results comparable to the previous section. In this section, the models were developed from scratch. As before, we tuned the hyperparameters for all our experiments. For this purpose, Optuna [34] was used to tune the hyperparameters of the models.

Using the PyTorch framework, a fully connected neural network comprising six hidden layers was developed. We optimized our model to identify the most effective hyperparameters for each target, including activation function, optimizer, criterion, learning rate, batch size, and dropout rate. The candidates for each parameter are listed in table S5. We employed Optuna and set the F1 score as the objective function for this optimization process. This step aimed to find the optimal set of parameters that would yield the highest F1 score.

In tables 5 and 6, we report the optimal hyperparameters for each dataset that resulted in the highest F1 score, along with the evaluation metrics obtained after fitting the model with those hyperparameters. Additionally, figure 8 illustrates the importance of each parameter, along with the ROC curves resulting from fitting the model with the best hyperparameters. As shown in figure 8(A), the most critical parameters for the Scaffold Quality dataset were dropout, optimizer, and learning rate; for printability, they were learning rate, optimizer, and number of epochs; and for Cell Response, they were dropout, learning rate, and activation function.

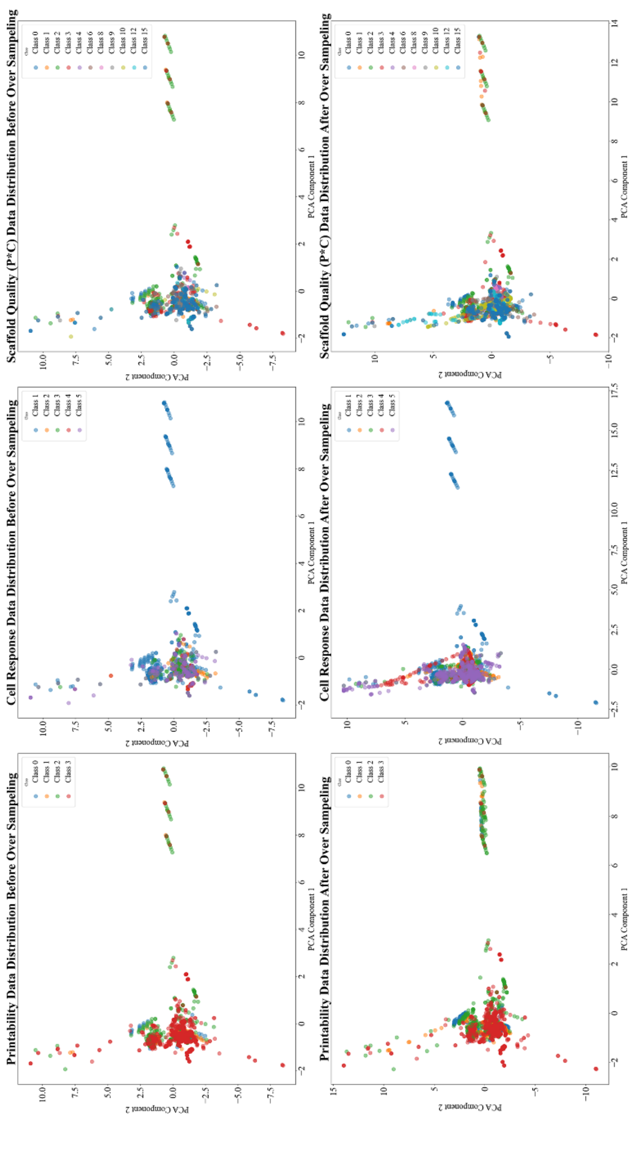

Figure 5. The visualization before and after running oversampling methods.

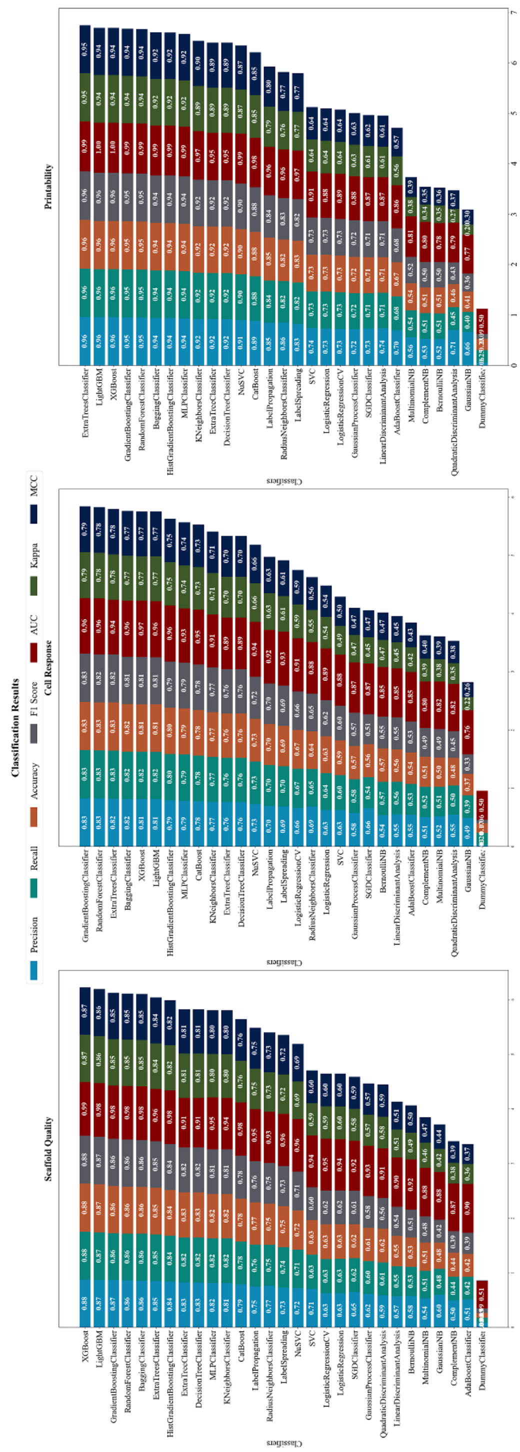

Figure 6. The results of training algorithms with tuned hyperparameters for predicting scaffold quality, cell response, and printability.

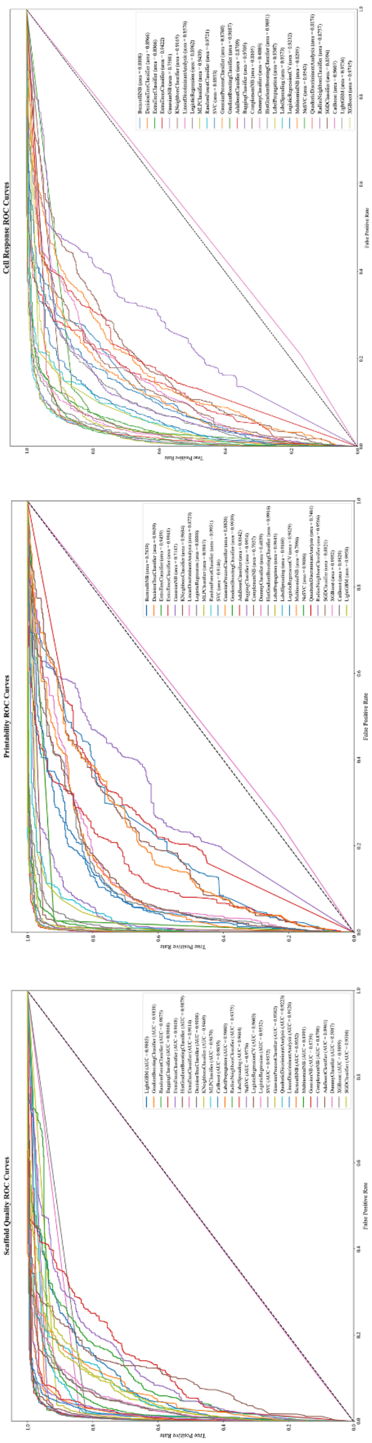

Figure 7. The ROC curves result from implementing different ML algorithms on the Scaffold Quality, Cell Response, and Printability datasets.

Table 5. Best hyperparameters for each dataset.

Dataset	Activation function	Optimizer	Criterion	Learning rate	Dropout	Batch size	Number of epochs
Scaffold Quality	PReLU	Adamax	CrossEntropyLoss	0.006 379	0.206202	64	1526
Printability	LeakyReLU	RMSprop	CrossEntropyLoss	0.000 202	0.161790	64	1100
Cell Response	ReLU6	Adamax	CrossEntropyLoss	0.001 846	0.306437	128	1759

Table 6. The results of training the neural networks using the best hyperparameters.

Target	Precision	Recall	Accuracy	F1 Score	AUC	Kappa	MCC
Scaffold Quality	0.804 212	0.806 114	0.806 957	0.801 738	0.971 435	0.787 603	0.788 238
Printability	0.934 206	0.930 83	0.932 692	0.931 773	0.983 921	0.910 115	0.910 552
Cell Response	0.794 882	0.796 451	0.793 046	0.790 56	0.958 981	0.741 288	0.743 82

4. Discussion

3D bioprinting is a technique used for fabricating scaffolds for tissue engineering. Bioink is an essential component of this process, consisting of one or more biomaterials loaded with cells or biomolecules such as drugs and growth factors. In this research, a dataset was built to develop a model for predicting the quality of scaffolds printed by 3D (bio)printing. This dataset included 60 major materials classified into five groups: natural polymers, synthetic polymers, small molecules, crosslinkers, and enzymes (figure 1(A) and table S1). Additionally, 49 cell lines (including six co-cultured systems) were used in cell-laden bioinks.

Selecting the proper biomaterials for the bioink is crucial as it directly impacts the final properties of the scaffold produced. The polymers' viscosity, hydration, mechanical stability, crosslinking mechanism, biocompatibility, and degradation properties are all important considerations for creating a high-quality bioink [42].

Natural polymers are a primary source for scaffold fabrication due to their numerous advantages. They can be efficiently used without harsh fabrication conditions or solvents, and their readily harvestable nature makes them invaluable [42]. Natural polymers dominate the majority of the materials used in the presented dataset. Five of the most used materials are alginate, gelatin, GelMA, HA, and nano/methyl cellulose (figure 1(A) and table S1).

Alginate is an anionic water-soluble polysaccharide. It is composed of two β-D-mannuronic acid (M) and α-L-glucuronic acid (G) residues, linked together with α-(1 $\rightarrow$ 4) glycosidic linkages [43]. Alginate has various properties, such as gelling capacity, cytocompatibility, water solubility, and low cost, that make it one of the most employed materials in 3D bioprinting [44]. It is readily printable and exhibits low toxicity and non-immunogenicity. The viscosity of alginate solutions depends on polymer concentration, molecular weight, cell phenotype, and density. For instance, lower alginate concentrations improve cellular responses while its mechanical stability significantly decreases. Thus, it is necessary to optimize the concentration [45]. In our dataset, the alginate concentrations varied from 0.25 to 20 w/v%, with an average of 3.59 $\pm$ 2.83 w/v% (table S1). To make alginate-based structures more stable, they can be ionically crosslinked using certain divalent cations, such as calcium dichloride ($CaCl_2$) [43]. As shown in figure 1(B), 508 scaffolds were printed using alginate and $CaCl_2$.

Single polymers typically lack adequate mechanical or biological properties to support hydrogels or short degradation time and swelling behavior. Therefore, adding second or third polymers to the bioink is necessary. Combining different biomaterials in the bioink may enhance the gelation ability of hydrogels, mechanical properties, biocompatibility, and cellular responses [43]. In 175 cases, alginate and gelatin were used together (figure 1(B)). Alginate lacks cell adhesion sites and forms soft hydrogels [46]. Adding a secondary biomaterial like gelatin can improve the alginate-based hydrogels' mechanical properties, cellular responses, and viscosity, thus enhancing their printability [47, 48]. On the other hand, the brittleness of gelatin hydrogels could be improved by adding alginate [49].

Other commonly used materials in the dataset are gelatin and its methacrylated derivative (GelMA). Gelatin is known for its biocompatibility, non-immunogenicity, high water absorption capacity, and rapid biodegradability [45]. Its non-immunogenicity, non-toxicity, and biodegradability led to its approval by FDA [46]. Gelatin can also retain collagen's bioactive sequences, providing a microenvironment for cellular activity such as adhesion, proliferation, and differentiation [50]. However, gelatin has limitations as it transforms from the gel state to the sol state at 30 °C–40 °C, hindering its printability in biological procedures. Its water resistance properties can be changed through chemical modification or crosslinking [46]. GelMA is a widely used modification of gelatin that can be photo-crosslinked using ultraviolet (UV) and water-soluble photoinitiators [45]. Additionally, GelMA is temperature-sensitive

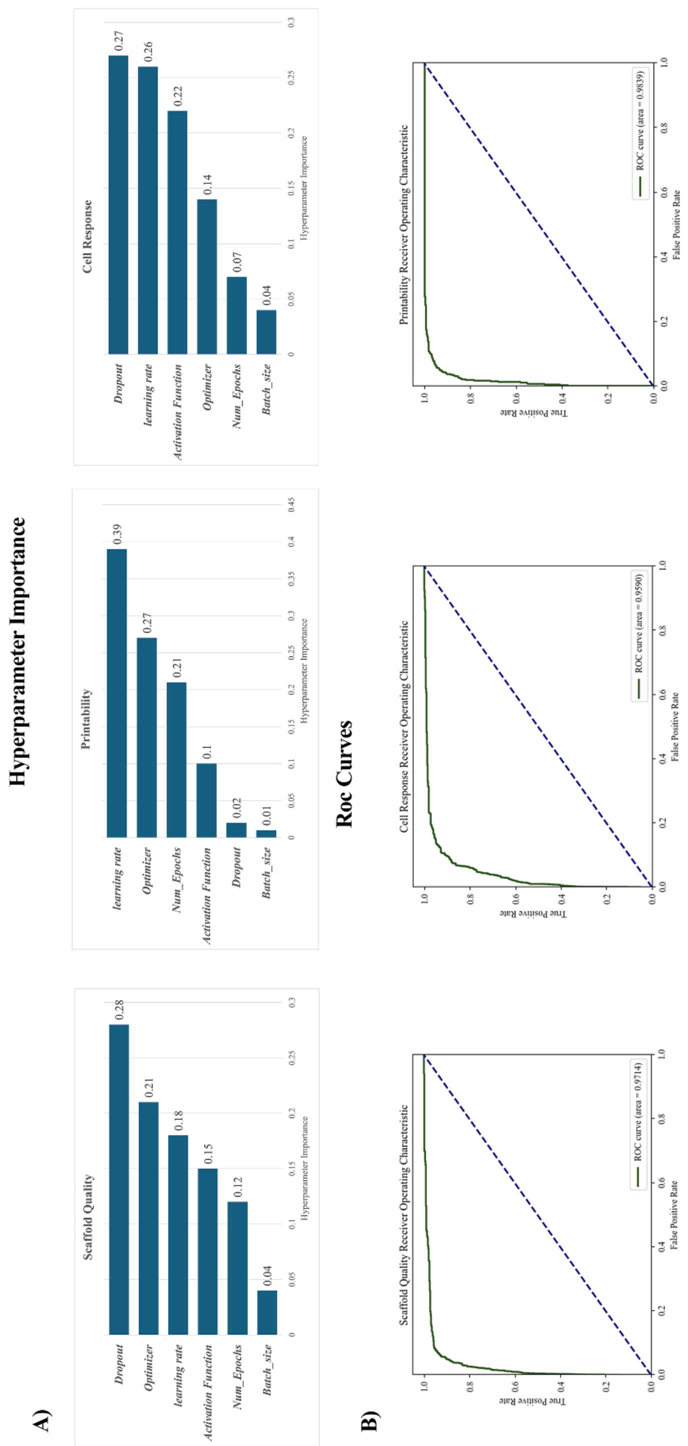

Figure 8. (A) Relative importance of hyperparameters in optimizing F1 score across datasets for Scaffold Quality, Printability, and Cell Response. (B) The roc curves resulted of fitting the best set of hyperparameters on scaffold quality, cell response and printability.

and can form physical gels at 0 °C in concentrations as low as 3 w/v% [51]. Our dataset includes studies that utilized gelatin concentrations ranging from 1 to 50 w/v% and 3–30 w/v% of GelMA, with an average of 12.72 ± 12.29 w/v% and 7.95 ± 4.41, respectively (table S1).

HA is a polysaccharide with several beneficial properties, such as biocompatibility, biodegradability, non-immunogenicity, and hydrophilicity, that can regulate cellular behaviors. The chemical structure of HA allows for chemical modifications, such as methacrylation, which enhances its mechanical strength, enables photocrosslinking upon UV irradiation, and slows its degradation rate [52]. Our dataset indicates that methacrylated HA (MeHA) was used as the primary component in 113 studies, with a concentration range of 2–5 w/v% and an average of 2.141 ± 0.621 w/v% (table S1). The material was mainly utilized at the lower end of its range, with a median of 2 w/v%. However, using HA or MeHA alone as bioinks presents challenges such as poor mechanical and rheological properties, cell attachment, and difficulty in photocrosslinking due to low methacrylation degrees. Therefore, MeHA is commonly paired with GelMA, an excellent standalone bioink that improves overall printability [53]. Conversely, adding MeHA to GelMA reduces its rapid degradation rate [54]. It is worth mentioning that in the developed dataset, 106 scaffolds combined MeHA with GelMA (figure 1(B)).

Finally, the fifth most used natural polymer was nano/methylcellulose, with 123 encounters in our dataset ranging from 0.25% to 12%. The mean of 3.62 ± 3.5 w/v% and median of 1.36 w/v% shows that most studies chose the lower percentage for scaffold production. Cellulose is the most abundant natural polymer with a crystalline structure, making it insoluble in water. Various functional groups can be added to its backbone to improve water solubility [43]. Methylcellulose is a cellulose ether with thermo-sensitive properties [55]. Another member of the cellulose family is cellulose nanofibril, which, based on the source, is categorized into three classes: microfibrillated, nanocrystalline, and bacterial cellulose [56]. 3D printing of nanocellulose-based bioinks is less common than nanocellulose-containing composites, possibly due to the lack of crosslinking sites on cellulose nanofibrils or their poor mechanical properties [57, 58]. Out of 113 3D-printed scaffolds, nano/methyl cellulose has been used alongside alginate in 70 cases (figure 1(B)). It has been shown that alginate/nanocellulose composite has higher storage and loss modulus, improving printability and ink fidelity compared to alginate solutions [59].

Synthetic polymers are another set of materials that are commonly used as bioinks. They are preferred over natural polymers because they lack batch-to-batch variation, immunogenicity, and impurity [60]. Surprisingly, fewer studies have utilized synthetic polymers in the presented dataset as their main components, possibly due to their inconvenient properties, such as slow degradability and inadequate biological properties like cell attachment. PEGTA, PCL, Pluronic P-123, Pluronic F127, and PEG-8-SH are five of the most used synthetic materials in this dataset (table S1).

Among the synthetic polymers, poly (ethylene glycol)-tetra-acrylate (PEGTA) is the most commonly used, with 35 encounters. It is often combined with alginate and gelatin in the range of 1–3 w/v%, with a mean of 2 ± 0.73 w/v% and median of 2 w/v%, respectively. PCL, another biodegradable synthetic polymer [61], was employed in 23 scaffolds with a minimum percentage of 50 and a maximum of 100. The median of 100 and mean of 95.56 ± 14.41 w/v% show that most studies utilized PCL alone. Pluronics are surfactants with two hydrophobic and one hydrophilic group, also known as poloxamers, in their molecular structure [62]. Pluronic P-123 and Pluronic F127 were used in 12 at concentrations ranging from 40%–60% to 40%–100%, respectively. Ultimately, an 8-arm poly(ethylene glycol) functionalized with thiol (PEG-8-SH) was used in all 8 instances at a concentration of 2.25% alongside gelatin and crosslinked with ruthenium and sodium persulfate in the presented dataset.

Crosslinkers are essential in 3D bioprinting as they can influence the final structure's mechanical stability and affect the encapsulated cells' cellular behavior [42]. Analyzing our dataset revealed that $CaCl_2$, Irgacure 2959, glutaraldehyde, LAP, and Eosin Y were the five most commonly used crosslinking agents (table S1). $CaCl_2$ was used in a broad concentration range of 0.02–1000 mM in 620 studies, with most studies using a concentration of 100 mM (mean 156.97 ± 178.21 mM and median of 100 mM). This crosslinker is widely used to crosslink alginate-based scaffolds, as 508 studies used alginate and $CaCl_2$ simultaneously (figure 1(B)). Irgacure 2959, a photoinitiator, was the second most commonly used crosslinker, with 223 encounters. It is primarily used to crosslink GelMA in a concentration range of 0.05–0.5 w/v%. In 215 scaffolds, GelMA was crosslinked with Irgacure 2959 (figure 1(B)). Glutaraldehyde and LAP were two other commonly used crosslinkers for these materials. On the other hand, Eosin Y was used in 22 individual scaffolds to crosslink alginate and alginate methacrylate alongside $CaCl_2$ in 23 individuals.

It is important to mention that two enzymes were also used in our dataset. Horse peroxidase radish (HRP) was used for enzymatic crosslinking in 13 scaffolds. It was often combined with HA (in 8 cases) and alginate (in 5 cases). Another enzyme, alginate lyase, was used in 7 studies along with alginate to speed up its degradation rate.

As seen in figure 1(D), considering the cell responses rating system, 47.7% of the dataset consisted of 3D bioprinted scaffolds. Bioinks used in 3D bioprinting consist of various components, with cells being one of the most crucial. While the incorporation of cells improves the functionality of printed structures and refines the tissue regeneration outcome, their proliferation, differentiation, and limited availability remain challenging. The cell source plays a significant role in their functionality. Primary cells, cell lines, and stem cells have been used in 3D bioprinting [63]. Chondrocytes were the most common cell used in this dataset. In total, 82 scaffolds were bioprinted using chondrocytes to promote chondrogenesis. Another common cell line was Bone marrow-derived mesenchymal stem cells (BMSCs), used primarily in bone and cartilage tissue engineering as they can differentiate into osteoblasts without needing growth factors [64]. In our dataset, 71 scaffolds were bioprinted with BMSCs (figure 2(A)). The dataset included three other commonly used cell lines: HepG2, a human hepatocellular carcinoma cell line; NIH3T3 fibroblasts; and mouse embryonic stem cells (mESCs). These three cell lines were observed 38, 36, and 28 times.

Regarding cell density (figure 2(B)), a broad spectrum ranging from 4500 to 30 $\times$ 10^6 cells ml^{-1} was utilized, with 1 $\times$ 10^6, 6 $\times$ 10^6, and 10 $\times$ 10^6 cells ml^{-1} being the most commonly used cell densities.

An analysis of physical and photo crosslinking duration indicates that most samples were not physically crosslinked. However, some samples were commonly crosslinked at 40 000 and 80 000 s. Regarding the photo crosslinking, most samples were crosslinked within 200 s (figure 2(C)).

The pressure of extrusion is a crucial printing parameter. According to the dataset, most printing was performed at low extrusion pressure levels (<200 kPa). However, printing in the range of 200–400 kPa was also common. Most printings within this range were done in the 200–300 kPa range. Printing was typically carried out with nozzle movement speeds below 50 mm s^{-1}, and most 3D printing was done with nozzles less than 500 micrometers in diameter. Moreover, the syringe temperature range in this dataset was mainly between 20 and 40 °C (figure 2(C)).

A 3D-printed scaffold's quality depends on the print quality and cell response. We determined the scaffold quantity metric, which is the multiplication of the printability score and cell response.

As previously noted, the performance of different ML algorithms is closely tied to dataset structure, and no single algorithm excels across all datasets. Accordingly, we tested various algorithms for each task, fine-tuning the hyperparameters to identify the optimal settings that yield the best performance for both our supervised and unsupervised tasks.

To ensure the integrity and quality of the dataset, we began with a thorough analysis to understand each dataset feature. It included a missing values imputation step, where we employed the Iterative Imputer with XGBoost regression as the estimator to predict and fill in missing values. This method leverages the relationships between existing data points to estimate missing values, thus maintaining the dataset's consistency and completeness.

Several ML techniques were applied, including supervised and unsupervised methods. Among the unsupervised techniques, we used clustering algorithms to uncover hidden patterns within the data. As shown in figure 3, the analyses indicated that the optimal number of clusters is five. Following this finding, we implemented 11 different clustering algorithms and evaluated them using the Silhouette, Davies-Bouldin, and Calinski-Harabasz scores. As discussed in the previous section, selecting the best-performing algorithm was based on finding a reasonable number of clusters, a high Silhouette Score and Calinski-Harabasz Score, and the lowest Davies-Bouldin Score. According to table S2, the KMeans algorithm demonstrated superior performance with five clusters.

The analysis of KMeans clustering revealed distinct characteristics for each cluster: Cluster 4 primarily consisted of physically crosslinked scaffolds, printed under high pressure, used larger nozzle diameters, and operated at higher syringe temperatures. Clusters 1 and 3 exhibited higher cell densities compared to others. In contrast, Cluster 2 predominantly utilized the photocrosslinking method and had a longer duration of exposure. Almost all of the 3D printed scaffolds were categorized into Cluster 0. By analyzing the clustering results in depth and comparing clusters, we identified which combinations of materials, cell lines, and printing conditions led to improved scaffold quality.

To deepen our understanding of the clusters, we mapped the clusters to our labels for cell response, printability, and scaffold quality. We discovered that most scaffolds in clusters 0 and 4 did not have cultured cells and were assigned a cell response label 1. Still, most received a printability label of 2, which means the printing was not optimized, resulting in a typical scaffold quality score of 2. It is important to note that if the scaffolds were 3D printed, their maximum score would be 3, so we had to judge their quality on a scale of 0–3. Most scaffolds in cluster 1 exhibited poor cell responses in the short term (receiving a cell response label of 2), yet they printed exceptionally well (all received a printability label of 3), leading to an average scaffold quality score of 6. Most scaffolds in cluster 2 were 3D printed and scored highly for printability. Lastly, most scaffolds in cluster 3 received printability labels of 9 and 12, indicating excellent printability and good short-term cell responses.

Given the distinct nature of our targets, classification algorithms were applied to the dataset. We employed cross-validation to ensure our models' robustness and mitigate the risk of overfitting. Cross-validation involves dividing the data into several subsets and training the model multiple times, using a different subset as the validation set and the remaining subsets as the training set. This process helps evaluate the model's performance more reliably and generalized. We split our dataset into training and test sets with a test ratio of 20%, ensuring that 80% of the data was used for training and 20% for testing. This independent test set provides an additional validation layer to evaluate the models on unseen data. Within the training set, we performed 10-fold cross-validation. In 10-fold cross-validation, the training data is randomly divided into 10 equal parts. Each part is used as a validation set, while the remaining 9 parts are used for training. This process is repeated 10 times, ensuring that every sample in the training dataset is used for training and validation. The performance metrics, including the F1 score, are averaged across all 10 folds, comprehensively evaluating the model's generalization ability. After fine-tuning the models using cross-validation, we evaluated the final models on the independent test set to obtain an unbiased estimate of their performance on truly unseen data. This rigorous process helps mitigate the risk of overfitting and ensures that the models are tested on various data subsets, enhancing their reliability.

However, due to the unbalanced data distribution across these classes, we first balanced our labels for all three targets using oversampling methods. Oversampling was crucial in balancing the dataset labels, which improved the training process and the overall performance of the ML algorithms. We employed several oversampling methods, including SMOTE, SVMSMOTE, ADASYN, RandomOverSampler, BorderlineSMOTE, and KMeansSMOTE. SMOTE generates synthetic samples for the minority class, thus ensuring that the classifier is not biased towards the majority class. After comparing the data distributions of the oversampled datasets with the original distribution, we chose SMOTE as it provided the most balanced and representative distribution for our analysis. Subsequently, we implemented 29 ML classification algorithms and fine-tuned their hyperparameters to identify the best-performing one for predicting printability, cell response, and scaffold quality. The highest F1 scores achieved were 0.96 for printability, 0.83 for cell response, and 0.88 for scaffold quality. Generally, tree-based algorithms demonstrated superior performance due to their ability to handle complex interactions between features and their robustness to overfitting when appropriately tuned.

We also developed a fully connected neural network using the PyTorch framework to predict these targets. This network features a sequence of layers strategically arranged to process the input data through a series of transformations. The model comprises six fully connected (dense) layers, implemented using PyTorch's nn.Linear module. The first dense layer maps the input features to a 128-dimensional space, followed by a series of progressively larger dense layers (128–256, 256–512), and then a symmetric decrease in layer sizes (512–256, 256–128), culminating in the final layer that maps the 128-dimensional representation to the num_classes output. Each layer is followed by a dropout layer, which is employed to mitigate overfitting—a common challenge in ML models—by randomly setting a fraction of the input units to zero at each update during training.

Each dense layer (except the final one) is followed by an activation function to introduce nonlinearity into the model and enable it to learn more complex representations. A flexible activation function approach allowed us to experiment with different activation functions such as ReLU, RReLU, LeakyReLU, SiLU, Tanh, etc. This architecture is driven by the need to balance model complexity and generalization. The symmetric expansion and contraction of layer sizes allow the model to expand first the feature space to capture intricate patterns and then condense this information into a form suitable for classification. Dropout regularization ensures the model does not overfit the training data, maintaining its ability to generalize to unseen data.

The training process involved splitting the dataset into training and testing sets using an 80-20 split. The features were scaled using MinMaxScaler to ensure faster convergence and converted to PyTorch tensors. The model was trained using various optimizers selected through hyperparameter tuning with Optuna. The hyperparameters tuned included the learning rate, dropout rate, batch size, number of epochs, and the choice of optimizer and activation function. The training loop ran for up to 2000 epochs, with batch sizes of 64, 128, or 256, and learning rates ranging from 1×10^{-5} to 1×10^{-1}.

Hyperparameter tuning was performed using Optuna, which optimized the model's performance over 150 trials. This process involved experimenting with different activation functions, dropout rates, learning rates, batch sizes, number of epochs, and optimizers to find the best combination of hyperparameters that maximized the F1 score. We tested various optimizers, including Adadelta, Adagrad, Adam, Adamax, etc. The role of the optimizer is to adjust the weights of the neural network to minimize the loss function during training. Different optimizers have various strategies for updating the weights, which can impact the convergence speed and the final performance of the model.

All seven evaluation metrics used for ML algorithms were calculated after finding the best set of hyperparameters. The highest F1 scores for printability, cell response, and scaffold quality predictions were 0.93, 0.79, and 0.80, respectively. The evaluation process involved calculating the F1 score for the model's test set predictions comparing true and predicted labels.

In comparison to ML algorithms, DL algorithms showed slightly lower performance. In general, ML algorithms demonstrated better performance on tabular data. However, the flexibility and potential of DL models make them a valuable tool for complex classification tasks, providing a strong baseline for further improvements and optimizations.

5. Conclusion

This study compiled a substantial dataset of 3D (bio)printed scaffolds for tissue engineering from previous works, encompassing a wide range of biomaterials, cell lines, and printing conditions. To the best of our knowledge, this is the most comprehensive open-source dataset in this field to date. Following data gathering, we conducted an in-depth analysis of our dataset characteristics to uncover detailed insights. Recognizing that each dataset has unique characteristics, we had to test and tune various algorithms to develop robust and reliable models that can perform well in real-world applications. Therefore, we implemented a range of supervised and unsupervised ML and DL algorithms to create models that can provide valuable information about each scaffold, determine its characteristics based on the cluster that it belongs to, and predict cell response, print quality, and overall quality of 3D (bio)printed scaffolds. This approach enables researchers to access a virtual lab, allowing them to investigate the quality of different scaffolds made from various materials and cells in a matter of seconds at no cost. Furthermore, we anticipate that this research will be a starting point for other AI applications in tissue engineering, like algorithmically designed scaffolds.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://github.com/saeedrafieyan/MLATE/. Data will be available from 01 August 2024.

Acknowledgments

We want to express our sincere appreciation to Fatemeh Norouzi, Maryam Adib, Maryam Dehghan-Niri, Behnaz Banimohamad-Shotorbani, and Zahra Ghasemi for their valuable collaborations.

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